



Cloudflare Radar 2024 Year in Review

A data-driven analysis of traffic, adoption, disruptions, and security signals (2024).

Audience: Technology executives and network infrastructure leaders

17% Global Traffic Growth and Starlink's 3.3x Surge

+17%

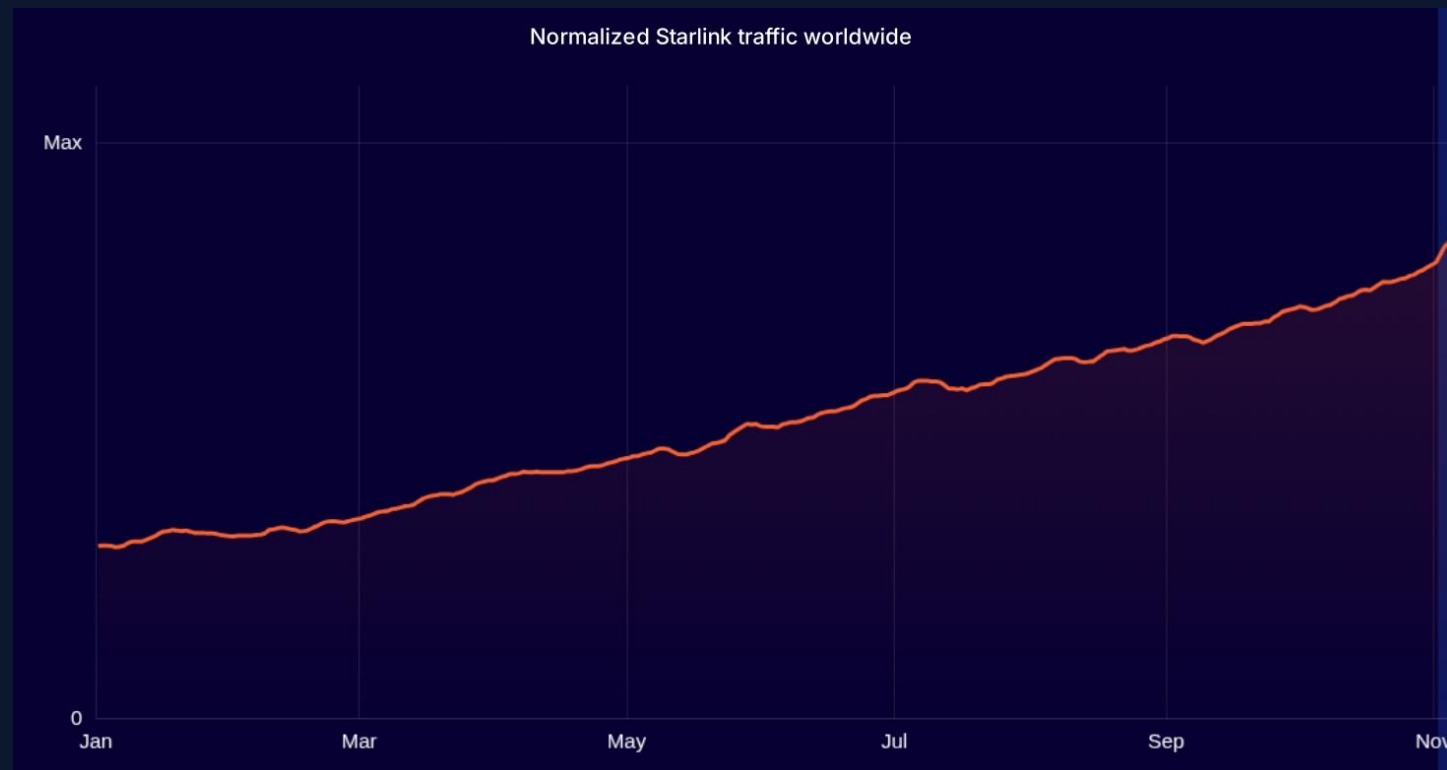
Global Internet traffic growth (YoY)
2024 vs 2023

3.3×

Starlink traffic growth
Normalized worldwide requests

Trend line

Method note
Starts mid-Jan to normalize



Starlink: 3.3× traffic growth in 2024 (normalized to peak request volume).

Google Leads Top 10 Services as Generative AI Rises

Google

#1 Service
2nd consecutive year

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI continued its rapid rise throughout 2024.

(Metaverse receded into the background.)

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

~2% → ~12–13%

Post-quantum encryption (TLS 1.3)

Jump after Chrome 124 default

21%

HTTP/3 adoption

Share of traffic using HTTP/3

Go (~12%)

Top API client language

Automated API requests

Post-quantum encrypted traffic worldwide



What changed

Chrome 124 enabled post-quantum key agreement by default on

Mozilla Firefox started rolling out PQ by default; Apple Safari beg

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

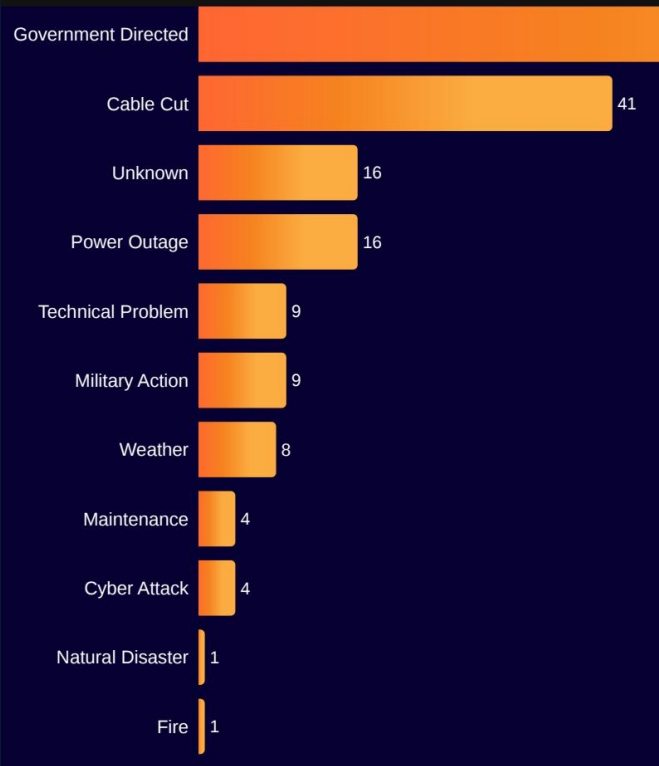
Major disruptions observed
Tracked via Radar Outage Center

Government-directed

Leading cause
116+ incidents

Cable / Power

Next causes
41 / 16



Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic mitigated
Of all traffic processed

Gambling & Games

Most-attacked sector
Industry category

68%

Bot concentration
From top 10 countries

Key security observations (2024)

- Cloudflare mitigated 6.5% of all traffic processed — threats remain pervasive at Internet scale.
- Gambling and Games was the most attacked industry category.
- Log4j (CVE-2021-44228) remains actively targeted more than two years after Dec 2021 disclosure.
- Email: 4.3% of emails processed were malicious; 43% of malicious emails contained a deceptive link.
- Email TLD signal: .bar originated the largest share of malicious and spam email observed.

Five Takeaways Shaping the 2025 Internet Outlook

● Sustained growth

Plan capacity for continued demand: +17% global traffic YoY in 2024.

● Connectivity fragility

Government-directed shutdowns were the leading cause of 225 major disruptions (

● Satellite acceleration

Starlink usage expanded rapidly: 3.3× traffic growth (normalized).

● Security persistence

6.5% of traffic mitigated; Log4j exploitation and email-borne threats persist.

● Crypto transition

Post-quantum encryption moved from ~2% to ~12–13% of TLS 1.3 traffic after Apr 17, 2024.

Supporting signals: HTTP/3 at 21% adoption; Go ~12% of automated API requests; 41% of traffic from mobile devices; IPv6 reached 28% of dual-stack traffic.